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Кафедра ИИТ

Лабораторная работа №7
По дисциплине: «Аппаратно-программное обеспечения ЭВМ и сетей»

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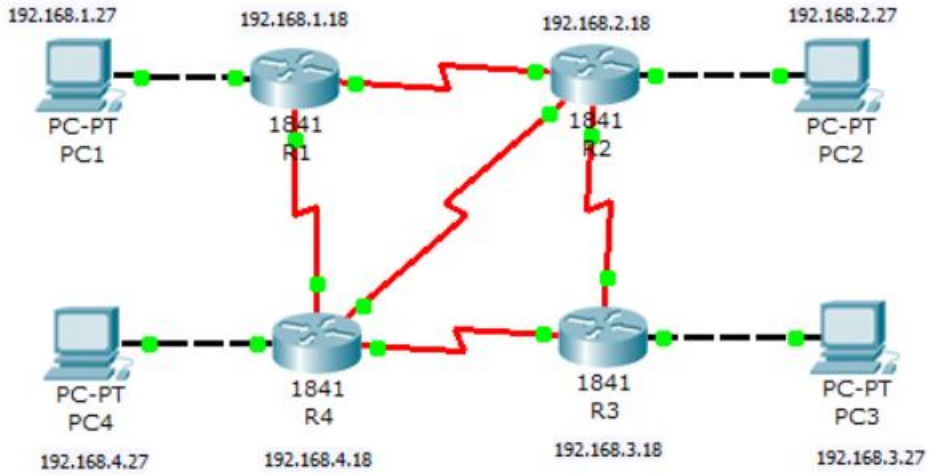
Брест 2026

Цель работы: Используя статическую маршрутизацию, обеспечить взаимодействие конечных устройств (PC1 и PC2). С помощью команды show и утилиты ping удостовериться, возможность взаимодействия конечных устройств обеспечена.

Вариант 17

Ход работы

Device	Interface	IP Address	Mask	Default Gateway
R1	Fa0/0	192.168.1.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.1	255.255.255.252	N/A
	Se0/0/1	10.0.0.5	255.255.255.252	N/A
R2	Fa0/0	192.168.2.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.2	255.255.255.252	N/A
	Se0/0/1	10.0.0.13	255.255.255.252	N/A
	Se0/1/0	10.0.0.9	255.255.255.252	N/A
R3	Fa0/0	192.168.3.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.14	255.255.255.252	N/A
	Se0/0/1	10.0.0.17	255.255.255.252	N/A
R4	Fa0/0	192.168.4.18	255.255.255.0	N/A
	Se0/0/0	10.0.0.6	255.255.255.252	N/A
	Se0/0/1	10.0.0.18	255.255.255.252	N/A
	Se0/1/0	10.0.0.10	255.255.255.252	N/A
PC1	N/A	192.168.1.27	255.255.255.0	192.168.1.18
PC2	N/A	192.168.2.27	255.255.255.0	192.168.2.18
PC3	N/A	192.168.3.27	255.255.255.0	192.168.3.18
PC4	N/A	192.168.4.27	255.255.255.0	192.168.4.18



PC1

Physical | Config | Desktop | Software/Services

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

IP Address.....: 192.168.1.27
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.1.18

PC>
```

PC2

Physical | Config | Desktop | Software/Services

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

IP Address.....: 192.168.2.27
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.2.18

PC>
```

PC3

Physical | Config | Desktop | Software/Services

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

IP Address.....: 192.168.3.27
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.3.18

PC>
```

PC4

Physical | Config | Desktop | Software/Services

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ipconfig

IP Address.....: 192.168.4.27
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.4.18

PC>
```

R1

Physical | Config | CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable secret cisco
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#interface fastethernet 0/0
R1(config-if)#ip address 192.168.1.18 255.255.255.0
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R1(config-if)#interface serial 0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
```

R1

Physical | Config | CLI

IOS Command Line Interface

```
o up

R1(config-if)#interface serial 0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#interface serial 0/0/1
R1(config-if)#ip address 10.0.0.5 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

R2

Physical Config CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable secret cisco
R2(config)#line console 0
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#exit
R2(config)#interface fastethernet 0/0
R2(config-if)#ip address 192.168.2.18 255.255.255.0
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R2(config-if)#interface serial 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown
```

R2

Physical Config CLI

IOS Command Line Interface

```
R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R2(config-if)#interface serial 0/0/1
R2(config-if)#ip address 10.0.0.13 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)#interface serial 0/1/0
R2(config-if)#ip address 10.0.0.9 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

R3

Physical Config CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable secret cisco
R3(config)#line console 0
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#exit
R3(config)#interface fastethernet 0/0
R3(config-if)#ip address 192.168.3.18 255.255.255.0
R3(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R3(config-if)#interface serial 0/0/0
R3(config-if)#ip address 10.0.0.14 255.255.255.252
R3(config-if)#no shutdown
```

R3

Physical Config CLI

IOS Command Line Interface

```
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

interface serial 0/0/1
R3(config-if)#interface serial 0/0/1
R3(config-if)#ip address 10.0.0.17 255.255.255.252
R3(config-if)#clock rate 64000
R3(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#
```

R4

Physical Config CLI

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R4
R4(config)#no ip domain-lookup
R4(config)#enable secret cisco
R4(config)#line console 0
R4(config-line)#password cisco
R4(config-line)#login
R4(config-line)#exit
R4(config)#interface fastethernet 0/0
R4(config-if)#ip address 192.168.4.18 255.255.255.0
R4(config-if)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state t
o up

R4(config-if)#interface serial 0/0/0
R4(config-if)#ip address 10.0.0.6 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

R4

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


R4(config-if)#interface serial 0/0/1
R4(config-if)#ip address 10.0.0.18 255.255.255.252
R4(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

R4(config-if)#interface serial 0/1/0
R4(config-if)#ip address 10.0.0.10 255.255.255.252
R4(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

R4(config-if)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console

R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        192.168.1.18    YES manual up                    up

FastEthernet0/1        unassigned      YES unset  administratively down down

Serial0/0/0            10.0.0.1        YES manual up                    up

Serial0/0/1            10.0.0.5        YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
```

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C       10.0.0.4 is directly connected, Serial0/0/1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R1#
```

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.2.18    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.2        YES manual up                    up
Serial0/0/1            10.0.0.13       YES manual up                    up
Serial0/1/0            10.0.0.9        YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C       10.0.0.8 is directly connected, Serial0/1/0
C       10.0.0.12 is directly connected, Serial0/0/1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
```

```
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.3.18    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.14       YES manual up                    up
Serial0/0/1            10.0.0.17       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.12 is directly connected, Serial0/0/0
C       10.0.0.16 is directly connected, Serial0/0/1
C    192.168.3.0/24 is directly connected, FastEthernet0/0
```

```
R4#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.4.18    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.6        YES manual up                    up
Serial0/0/1            10.0.0.18       YES manual up                    up
Serial0/1/0            10.0.0.10       YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
R4#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.0.0.4 is directly connected, Serial0/0/0
C       10.0.0.8 is directly connected, Serial0/1/0
C       10.0.0.16 is directly connected, Serial0/0/1
C    192.168.4.0/24 is directly connected, FastEthernet0/0
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
R1(config)#ip route 192.168.4.0 255.255.255.0 10.0.0.6
R1(config)#ip route 10.0.0.8 255.255.255.252 10.0.0.2
R1(config)#ip route 10.0.0.12 255.255.255.252 10.0.0.2
R1(config)#ip route 10.0.0.16 255.255.255.252 10.0.0.6
R1(config)#end
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.14
R2(config)#ip route 192.168.4.0 255.255.255.0 10.0.0.10
R2(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.1
R2(config)#ip route 10.0.0.16 255.255.255.252 10.0.0.14
R2(config)#end
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.13
R3(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.13
R3(config)#ip route 192.168.4.0 255.255.255.0 10.0.0.18
R3(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.13
R3(config)#ip route 10.0.0.4 255.255.255.252 10.0.0.18
R3(config)#ip route 10.0.0.8 255.255.255.252 10.0.0.13
R3(config)#end
```

```
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.5
R4(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.9
R4(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.17
R4(config)#ip route 10.0.0.0 255.255.255.252 10.0.0.5
R4(config)#ip route 10.0.0.12 255.255.255.252 10.0.0.9
R4(config)#end
```

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 5 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
C       10.0.0.4 is directly connected, Serial0/0/1
S       10.0.0.8 [1/0] via 10.0.0.2
S       10.0.0.12 [1/0] via 10.0.0.2
S       10.0.0.16 [1/0] via 10.0.0.6
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.0.2
S    192.168.3.0/24 [1/0] via 10.0.0.2
S    192.168.4.0/24 [1/0] via 10.0.0.6
```

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 5 subnets
C       10.0.0.0 is directly connected, Serial0/0/0
S       10.0.0.4 [1/0] via 10.0.0.1
C       10.0.0.8 is directly connected, Serial0/1/0
C       10.0.0.12 is directly connected, Serial0/0/1
S       10.0.0.16 [1/0] via 10.0.0.14
S    192.168.1.0/24 [1/0] via 10.0.0.1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
S    192.168.3.0/24 [1/0] via 10.0.0.14
S    192.168.4.0/24 [1/0] via 10.0.0.10
```

```
R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 5 subnets
S       10.0.0.0 [1/0] via 10.0.0.13
S       10.0.0.4 [1/0] via 10.0.0.18
S       10.0.0.8 [1/0] via 10.0.0.13
C       10.0.0.12 is directly connected, Serial0/0/0
C       10.0.0.16 is directly connected, Serial0/0/1
S    192.168.1.0/24 [1/0] via 10.0.0.13
S    192.168.2.0/24 [1/0] via 10.0.0.13
C    192.168.3.0/24 is directly connected, FastEthernet0/0
S    192.168.4.0/24 [1/0] via 10.0.0.18
```

```
R4#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 5 subnets
S       10.0.0.0 [1/0] via 10.0.0.5
C       10.0.0.4 is directly connected, Serial0/0/0
C       10.0.0.8 is directly connected, Serial0/1/0
S       10.0.0.12 [1/0] via 10.0.0.9
C       10.0.0.16 is directly connected, Serial0/0/1
S    192.168.1.0/24 [1/0] via 10.0.0.5
S    192.168.2.0/24 [1/0] via 10.0.0.9
S    192.168.3.0/24 [1/0] via 10.0.0.17
C    192.168.4.0/24 is directly connected, FastEthernet0/0
```

PC1

Physical | Config | Desktop | Software/Services

Command Prompt

```
Pinging 192.168.2.27 with 32 bytes of data:

Reply from 192.168.2.27: bytes=32 time=94ms TTL=126
Reply from 192.168.2.27: bytes=32 time=93ms TTL=126
Reply from 192.168.2.27: bytes=32 time=92ms TTL=126
Reply from 192.168.2.27: bytes=32 time=76ms TTL=126

Ping statistics for 192.168.2.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 76ms, Maximum = 94ms, Average = 88ms
```

PC2

Physical | Config | Desktop | Software/Services

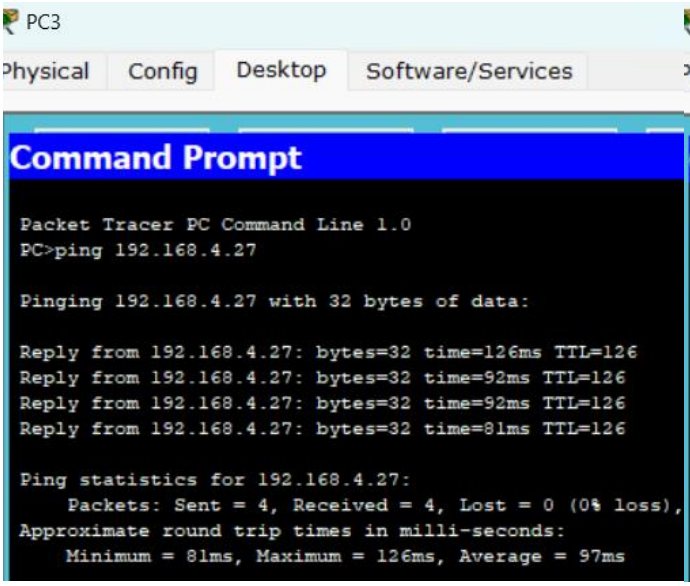
Command Prompt

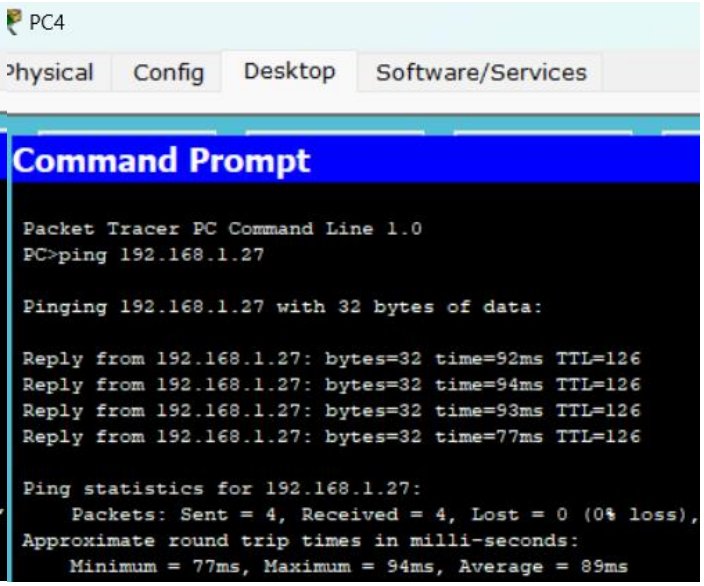
```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.4.27

Pinging 192.168.4.27 with 32 bytes of data:

Reply from 192.168.4.27: bytes=32 time=90ms TTL=126
Reply from 192.168.4.27: bytes=32 time=96ms TTL=126
Reply from 192.168.4.27: bytes=32 time=91ms TTL=126
Reply from 192.168.4.27: bytes=32 time=93ms TTL=126

Ping statistics for 192.168.4.27:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 90ms, Maximum = 96ms, Average = 92ms
```





Вывод: я изучила возможности и порядок применения пакета Cisco Packet Tracer. Приобрела навыки по начальному конфигурированию маршрутизаторов.